MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

1. **PROGRAMME** : Bachelor of Science
2. **SUBJECT** : Information and Communication Technology
3. **LEVEL OF STUDY** : 1
4. **COURSE TITLE** : Introduction to Programming with C
5. **COURSE CODE** : BICT 1102
6. **DURATION** : 16 Weeks
7. **PRESENTED TO** : Senate
8. **PRESENTED BY** : Dean of Science, Technology and Innovation
9. **LECTURE HRS PER WEEK** : 4
10. **TUTORIAL HRS PER WEEK** : 2
11. **PRACTICAL HOURS** : 1
12. **INDEPENDENT LEARNING HRS PER WEEK** : 9
13. **TOTAL COURSE CREDITS** : 13
14. **PRE-REQUISITE COURSE CODES(S)**: BICT 1102
15. **CO-REQUISITE COURSE CODES(S)**: None
16. **DELIVERY METHODS:**
    - 16.1 **Mode of Delivery** : Face- to Face
    - 16.2 **Teaching Methods** : Lectures and tutorials
17. **ASSESSMENT METHODS:** Minimum of two continuous tests
One end of semester examination
18. **ASSESSMENT WEIGHTING** : 40% Continuous
60% End of semester
19. **COURSE DESCRIPTION** : The course is designed to provide students with a thorough foundation in the key concepts, techniques and methods that have emerged over time in computer programming. To help students acquire knowledge and skills in C programming and apply in other structured programming languages.
20. **AIMS OF THE COURSE:**
    - To provide students with a thorough foundation in the key concepts, techniques and methods that have emerged over time in computer programming

- To help students acquire knowledge and skills in C programming and apply them in other programming languages

21. **LEARNING OUTCOMES**:

A successful learner from this course will be able to:

- describe developments in programming languages and reasons for their evolution
- apply appropriate ways to control and manipulate data.
- break down data into structures appropriate for the problem to be addressed
- formulate data into structures appropriate for the problem to be addressed
- utilize a range of common programming concepts as building blocks in the creation of structured programs
- define code requirements from the models
- apply a range of techniques to modeling systems as objects
- Apply a range of testing approaches to check the correctness of programming code and explain other evolving ways of designing and building programs.
- describe the importance of documentation in software development

- Write a program in C to solve specific problem
- Appreciate the modular nature of a computer program

22. TOPICS OF STUDY:

History of programming

- Describing Programming language
- Language Models
- Chronology of Programming
- Object Oriented Concepts
- Classes
- Encapsulation (or Information Hiding) and Abstraction
- Messages and Operations
- Relationships
- Polymorphism

Introduction to C

- Characteristics of C
- Compilers and Interpreters
- Lexical conventions

The C Library

- # include files
- Keywords
- Constants
- Other useful tools
- Basic Data Types (int, float, double, char)
- More on floating points
- Modifying data types (long, short, signed, unsigned)

Complex Data Types

- Union
- Struct
- Void

Variable declarations/initialisations

- Variable definition
- Variable declarations
- Storage classes and scope of a variable
- Variable initialisation

Operators

- Operators by category
- Operator precedence and associativity
- Order of evaluation
- Functions

Standards and Programming Style

- Modular code design
- Writing a C program

Input and Output

- EOF
- Streams
- Input functions
- Output functions
- Format symbols

Control Flow and Program Loops

- The while and do while loops
- The if statement
- The for statement
- Control modifiers

Data Structures

- Arrays
- Multi-dimensional arrays
- Structures
- Unions
- Declaring a structure
- Initialising a structure
- Referencing a struct or union member

Pointers

- Declaring a pointer
- Referencing a pointer
- Pointer differences
- Arrays as pointers

- Pointers in function arguments
- Command Line Arguments
- Complex Declarations
- Reading
- Writing
- Typedef

Handling File I/O

- File pointers and descriptors
- Closing an open file
- Opening files
- String Manipulations
- Memory Management
- Low level file I/O
- The malloc family of functions

23. PRESCRIBED TEXTS:

Dietel, H.M. and Dietel, P.J. (2016). *C: How to program*, (8th ed.). New Jersey: Pearson.

24. RECOMMENDED TEXTS

Gookin, D. (2014). *Beginning Programming with C for Dummies*. New Jersey: John Wiley & Sons, Inc

Bennett, S., Wixom, B. H, Tegarden, D. P. (2012). *Object oriented Systems analysis design* (5th ed.). McGraw-Hill.

Rambaugh, J, Jacobson, I., Booch, G. (2004). *The Unified Language Reference Manua*l (2nd ed.). Boston: Addison Wesley Professional.